FOR EDITH · PAEDIATRICS & CHILD HEALTH JUNIOR CLERKSHIP

Common Neonatal Disorders

Tutorial | Week 5 · Measles, vaccines & the newborn | Neonatology | Dr Namugga

In the first 28 days the commonest problems on a Ugandan newborn ward are sepsis, jaundice, hypoglycaemia, hypothermia, local infections of the cord, eyes and skin, and birth injuries. Recognise the danger signs early, start ampicillin plus gentamicin in weight- and age-based doses for possible serious bacterial infection, treat jaundice against bilirubin thresholds, and keep every baby warm, fed and glycaemic.

Must know before the ward round

1. Neonatal danger signs (WHO/UCG) - not feeding well, convulsions, drowsy/unconscious or movement only when stimulated, RR 60 or more, severe chest indrawing, grunting, temperature above 37.5 or below 35.5 C, central cyanosis.
2. Possible serious bacterial infection - ampicillin 50 mg/kg IV/IM 12-hourly in week 1 (8-hourly weeks 2-4) PLUS gentamicin 5 mg/kg once daily in week 1 (3 mg/kg LBW per WHO; UCG 4 mg/kg preterm) and 7.5 mg/kg once daily in weeks 2-4.
3. Duration - culture-negative and well at 48 h - stop; sepsis 7-10 days (UCG 10 days); meningitis 21 days. If no improvement in 48-72 h, change to ceftriaxone 100 mg/kg once daily (UCG) or per culture.
4. Staphylococcal risk (omphalitis with flare, many pustules, abscess) - cloxacillin 50 mg/kg IV/IM 6-hourly plus gentamicin (UCG 2023).
5. Jaundice in the first 24 h is ALWAYS pathological - measure bilirubin within 2 h (NICE) and think haemolysis (ABO, Rh, G6PD) or sepsis.
6. Yellow palms and soles at any age, or any jaundice on day 1 = severe jaundice - refer/admit for bilirubin and phototherapy (UCG/IMNCI).
7. NICE threshold table (38 weeks or more) - phototherapy at 100 umol/L at birth rising to 200 at 24 h, 250 at 48 h, 300 at 72 h and 350 from 96 h; exchange at 100 (birth), 300 (24 h) and 450 from 42 h.
8. Prolonged jaundice - more than 14 days in term, more than 21 days in preterm babies; check conjugated bilirubin and stool colour (pale stools = biliary atresia, operate before 60 days).
9. Hypoglycaemia - treat below 2.6 mmol/L (Uganda practice; WHO 2.2, NICE 2.0) with feeds if well, or 10% dextrose 2 mL/kg IV then infusion if symptomatic or not feeding; recheck in 30 min.
10. Hypothermia - normal 36.5-37.5 C; moderate 32-35.9; severe below 32. Rewarm with skin-to-skin/KMC or incubator, check glucose, look for sepsis.
11. Gonococcal ophthalmia neonatorum - ceftriaxone 125 mg IM once (WHO 50 mg/kg, max 125 mg) PLUS azithromycin 20 mg/kg daily for 3 days (UCG), saline eye washes, treat both parents.
12. Cord care in Uganda - chlorhexidine 7.1% digluconate gel daily for 7 days, nothing else on the stump; local umbilical infection - clean, gentian violet 0.5% three times daily, oral amoxicillin, review in 2 days.

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In a nutshell

The newborn has a narrow margin of safety: little energy reserve, immature immunity and a big surface area. The same few problems cause most admissions and deaths after the first day: **sepsis**, **jaundice**, **hypoglycaemia**, **hypothermia**, **local infections** (cord, eyes, skin) and **birth injuries**, often together in the same baby.

Why it matters in Uganda: **infection** is one of the three leading causes of neonatal death (with prematurity and asphyxia). Severe jaundice causes **kernicterus**, which still leaves Ugandan children with athetoid cerebral palsy and deafness. Hypothermia and hypoglycaemia are cheap to prevent and deadly when missed.

The approach is the **IMNCI** one: look for **danger signs** first, classify, treat (often first-dose antibiotics and glucose before referral), keep warm, keep feeding, and counsel the mother on when to return.

> **CLINICAL PEARL**
> The one thing never to forget: **a newborn shows almost every serious illness the same way** (poor feeding, lethargy, temperature instability, fast breathing, jaundice). Check **temperature and glucose**, and if any danger sign is present, **start antibiotics** after a blood culture (where available) and **do not wait** for results.

Definitions & classification

Term	Definition
Neonatal period	First **28 days** of life (early neonatal = first 7 days)
Early-onset sepsis (EOS)	Onset in the first **72 hours** (some use 7 days); usually maternal organisms acquired in labour
Late-onset sepsis (LOS)	Onset after **72 hours**; community or hospital-acquired (hands, equipment, cord)
Possible serious bacterial infection (PSBI)	IMNCI classification for a young infant with any danger sign; treat as sepsis
Physiological jaundice	Appears **after 24 h**, peaks day 3-5 (term), mild, baby well, gone by **14 days** (term) / 21 days (preterm)
Pathological jaundice	**First 24 h**, rapidly rising, above treatment thresholds, **palms/soles yellow**, baby unwell, conjugated, or prolonged
Neonatal hypoglycaemia	Blood glucose **below 2.6 mmol/L** (operational threshold in Ugandan units; WHO Pocket Book 2.2; NICE/BAPM 2.0)
Hypothermia	Axillary **below 36.5 C**: cold stress 36.0-36.4, moderate 32.0-35.9, **severe below 32**
Omphalitis	Infection of the umbilical stump; "**local**" (red, pus) vs **periumbilical flare** (redness spreading to the abdominal wall = serious)
Ophthalmia neonatorum	Conjunctivitis in the first **28-30 days** of life, classically gonococcal or chlamydial

Epidemiology & Uganda context

- Sepsis in Ugandan and other sub-Saharan newborn units is commonly caused by **Klebsiella pneumoniae, Staphylococcus aureus, E. coli** and other Gram-negatives; **Group B Streptococcus** is less often isolated

than in HICs but does occur. **Resistance to ampicillin and gentamicin is common** in hospital-acquired infections, which is why second-line drugs and cultures matter.

- **Hand hygiene, chlorhexidine cord care, exclusive breastfeeding and KMC** are the main prevention tools. Harmful traditional practices (cow dung, ash or herbs on the cord; "sucking" the baby's mouth; prelacteal feeds) still occur.
- **Jaundice**: G6PD deficiency is common in Uganda, as are ABO incompatibility, sepsis and prematurity; Rh disease is less common because few Ugandan women are Rh-negative, but it is severe when it occurs. Many babies are discharged within 24 h, so **jaundice is often first seen at home** and presents late.
- **Facilities**: HC III/IV can give first-dose IM antibiotics, glucose/feeds, KMC and refer; phototherapy and bilirubin measurement are mainly at **general/regional and national referral hospitals** (and private-not-for-profit hospitals such as Mengo, Nsambya, Lubaga); **exchange transfusion** at a few referral centres.

Pathophysiology

Why newborns get infected easily: low complement and neutrophil reserves, little IgM, IgG mainly from the mother (crosses after 32 weeks, so preterms lack it), thin skin, an open umbilical wound, and exposure to maternal organisms in labour (prolonged ROM, chorioamnionitis). Sepsis spreads quickly to the meninges (up to a quarter of septic newborns have meningitis).

Why newborns get jaundiced: high red cell mass with a short lifespan (70-90 days) → a large bilirubin load; immature **UDP-glucuronyl transferase** in the liver; more **enterohepatic circulation** (gut beta-glucuronidase, sterile gut, slow passage of meconium). Unconjugated bilirubin is fat-soluble and, when very high (or with acidosis, sepsis, prematurity, low albumin, drugs that displace it such as ceftriaxone), crosses the blood-brain barrier and stains the **basal ganglia → acute bilirubin encephalopathy** and **kernicterus**.

Why newborns get hypoglycaemic: after the cord is cut, glucose supply stops; the baby relies on glycogen (small in preterm/SGA), gluconeogenesis and ketones from milk. Risk is greatest in **preterm, SGA, infants of diabetic mothers (hyperinsulinism), asphyxia, sepsis, hypothermia and delayed feeding**.

Why newborns get cold: large surface area, wet skin at birth (evaporation), little fat, cannot shiver (rely on brown fat, which uses oxygen and glucose). Cold → hypoglycaemia, acidosis, apnoea, sepsis, death.

Clinical acumen

Focused history

Question	Why it matters
Age in hours/days	EOS vs LOS; jaundice timing (day 1 = pathological)
Feeding: breastfeeding? how often? last good feed? other fluids/prelacteal feeds?	Poor feeding is the most common sign of illness; cause of hypoglycaemia and dehydration
Maternal fever, ROM over 18 h, foul liquor, UTI in labour	EOS risk; give antibiotics
Gestation and birth weight	Preterm/LBW: sepsis, jaundice and hypoglycaemia risk; lower bilirubin thresholds
Mother's blood group (and Rh), previous baby with jaundice/exchange	ABO/Rh haemolysis
Family history of G6PD deficiency or jaundice; exposure to naphthalene/herbs	Haemolysis triggers
Stool and urine colour	Pale stools/dark urine = conjugated (cholestatic) jaundice

Question	Why it matters
Cord care: what was applied?	Omphalitis, tetanus
Eye discharge onset	Day 2-5 gonococcal; day 5-14 chlamydial
Delivery: vacuum, breech, shoulder dystocia, big baby	Birth injuries
Maternal HIV (last VL), syphilis (RPR), HBsAg, TB	Prophylaxis and congenital infection
Immunisation: BCG, OPV0 (HepB birth dose)	Missed doses; BCG ulcer vs abscess

Examination

- **Vital signs**: temperature (axillary), RR (count 60 s; repeat if 60 or more), HR, SpO2, CRT, weight (compare with birth weight).
- **Danger signs** (WHO/UCG): not feeding, convulsions, lethargy/unconscious, movement only when stimulated, RR 60 or more, severe indrawing, grunting, temperature above 37.5 or below 35.5 C, central cyanosis.
- **Jaundice**: press the skin in good daylight; note how far down it extends (face → chest → abdomen → limbs → **palms and soles**). Look for pallor, hepatosplenomegaly, cephalhaematoma, bruising.
- **Neurology** (in jaundice): lethargy, poor suck, hypotonia → hypertonia, retrocollis, opisthotonus, high-pitched cry (acute bilirubin encephalopathy).
- **Cord**: redness, pus, smell, **flare** spreading onto the abdominal wall, bleeding.
- **Eyes**: swelling, pus (copious = gonococcal), corneal haze.
- **Skin**: pustules (count: fewer than 10 vs 10 or more), bullae, abscess, cellulitis, peeling.
- **Mouth**: thrush (white patches that do not wipe off).
- **Bones and nerves**: arm movement (Erb's palsy), clavicle crepitus, scalp swellings.

Red flags

RED FLAG

- Any **danger sign** (above) = PSBI: first-dose antibiotics and urgent referral/admission
- **Jaundice on day 1**, **yellow palms/soles**, jaundice with **lethargy, poor suck, high-pitched cry, arching**
- **Pale stools / dark urine** with jaundice (biliary atresia)
- **Glucose below 2.6 mmol/L** with symptoms, or persisting despite feeds
- **Temperature below 35.5 C** or above 37.5 C
- **Periumbilical flare** or bleeding from the cord
- **Copious purulent eye discharge** with lid swelling in the first week
- **Bulging fontanelle**, convulsions (meningitis)
- **Bilious vomiting** or abdominal distension (surgical emergency/NEC)

Thinking like a clinician

- **Sepsis looks like everything.** Hypothermia, hypoglycaemia, jaundice, apnoea, poor feeding and "just not right" can each be the only sign. Lower your threshold to treat.
- **Fever in a newborn is serious, but so is a low temperature.** Hypothermia is more common than fever in neonatal sepsis, especially in preterms.
- **Early jaundice = haemolysis or sepsis; late or prolonged jaundice = breast milk, hypothyroidism, sepsis/UTI, or cholestasis.** Always look at the stool colour.
- **A jaundiced baby who is sleepy and feeding poorly is either septic or developing bilirubin encephalopathy.** Both are emergencies.

- **Do not give ceftriaxone to a deeply jaundiced or preterm neonate** if an alternative exists (it displaces bilirubin from albumin), and never mix it with IV calcium.
- **A cold baby is a hypoglycaemic baby until proven otherwise**, and a hypoglycaemic baby may be septic.
- **Not all skin spots are infection**: erythema toxicum (blotchy red with white/yellow centres, day 2-3, baby well), milia, miliaria and Mongolian spots are benign. Pustules with surrounding redness in a baby with other signs are not.

Clinical workup

Step 0: First 5 minutes

1. **ABC**: open airway; oxygen by nasal prongs if SpO2 below 90% or cyanosed/severe distress; bag-mask if apnoeic or gasping.
2. **Temperature**: if below 36.5 C start rewarming (skin-to-skin, incubator/radiant warmer); if above 37.5 C undress/adjust environment (no paracetamol).
3. **Glucose** (glucometer): below 2.6 mmol/L → **10% dextrose 2 mL/kg IV** (or EBM/10% dextrose by NGT if no IV).
4. **Danger sign present?** → IV/IO access, **blood culture**, **ampicillin + gentamicin** now; **phenobarbital 20 mg/kg** if convulsing.
5. **Shock?** (CRT above 3 s, weak pulses, HR above 180) → **normal saline 10 mL/kg** over 10-20 min, reassess.
6. **Vitamin K** if not given.
7. Call for senior help; explain to the mother.

Step 1: Focused history

- [ ] Age in hours/days and onset of the problem: EOS vs LOS; jaundice timing.
- [ ] Feeding and urine/stool output: dehydration, hypoglycaemia, cholestasis.
- [ ] Labour risk factors (ROM over 18 h, maternal fever, foul liquor).
- [ ] Gestation and birth weight; percentage weight change.
- [ ] Maternal blood group/Rh, previous jaundiced baby, family G6PD.
- [ ] What was applied to the cord or eyes; home remedies.
- [ ] Delivery mode and trauma.
- [ ] Maternal HIV (VL), syphilis, HBsAg; immunisations given.

Step 2: Focused examination

- General: activity, colour (pink/pale/yellow/blue), cry, posture.
- Vitals, weight, CRT; signs of dehydration (sunken fontanelle, dry mucosa).
- Jaundice extent and palms/soles; liver and spleen.
- Neurology: tone, suck, fontanelle.
- Cord, eyes, skin, mouth.
- Chest (RR, indrawing, grunting) and abdomen (distension).
- Limbs and scalp for birth injury.

Step 3: Bedside tests

Test	What it tells you
Glucose	Hypoglycaemia (below 2.6) or stress hyperglycaemia (sepsis)
Temperature	Hypothermia grade or fever

Test	What it tells you
SpO2	Hypoxaemia (pneumonia, sepsis, CHD)
Transcutaneous bilirubin (TcB)	Screening in babies of 35 weeks or more after 24 h (NICE); confirm with serum if high or day 1
Weight	Weight loss above 10% suggests poor intake/dehydration
HemoCue Hb	Pallor, haemolysis, bleeding
Urine dipstick (bag/clean catch)	UTI in prolonged jaundice or LOS
Malaria RDT/film	Fever in a malaria-endemic area (congenital/neonatal malaria is rare but possible)

Step 4: Laboratory & imaging

Test	When	Expected / abnormal finding	How it changes management	Availability
Blood culture	Every suspected sepsis, **before antibiotics**	Organism and sensitivity	Choice and duration; stop at 48 h if negative and well	Regional/national referral, PNFP hospitals
FBC	Sepsis, jaundice, pallor	WBC below 5 or above 20, immature neutrophils, platelets below 100; Hb low and reticulocytes high in haemolysis	Supports sepsis; transfusion; haemolysis work-up	HC IV and above
CRP	Sepsis	Rising at 18-24 h; normal values at 0 and 18-24 h help stop antibiotics	Duration	Referral hospitals
Lumbar puncture (CSF)	Sepsis with seizures, bulging fontanelle, lethargy, or positive culture	WBC above about 20/mm3, low glucose, high protein, organisms	**21 days** of IV antibiotics	Hospital
Total serum bilirubin (TSB)	Jaundice day 1, palms/soles yellow, TcB high, preterm	Plot on threshold chart by age in hours	Phototherapy/exchange	General/regional referral
Conjugated (direct) bilirubin	Jaundice beyond 14 days (term)/21 days (preterm), pale stools	**Above 25 umol/L** (NICE) = cholestasis	Urgent referral for biliary atresia	Referral hospitals
Blood group, Rh and DAT (Coombs) of baby and mother	Early jaundice	ABO (mother O, baby A/B) or Rh incompatibility; DAT positive	Lower thresholds, IVIG/exchange	Referral hospitals
G6PD screen	Unexplained haemolysis, family history, male	Deficient	Avoid triggers; counsel family	Limited (tertiary/private)
Thyroid function	Prolonged jaundice	Low T4, high TSH	Levothyroxine	Referral hospitals
Eye swab: Gram stain and culture	Purulent conjunctivitis	Gram-negative intracellular diplococci = gonococcus	Confirms; contact tracing	Referral hospitals
Pus swab	Abscess, omphalitis with flare	S. aureus (MRSA?)	Antibiotic choice	Referral hospitals

Test	When	Expected / abnormal finding	How it changes management	Availability
X-ray	Suspected fracture (clavicle, humerus), pneumonia	Fracture, consolidation	Immobilisation; pneumonia treatment	Most hospitals

Step 5: Putting it together

Model summary line: "A **6-day-old** term male (birth weight 3.2 kg, now 2.9 kg, **9% loss**) born at home, brought with **2 days of poor feeding and fever 38.4 C**, with a **red umbilicus with pus and redness spreading 2 cm onto the abdominal wall**, RR 64, glucose 2.3 mmol/L, jaundice to the abdomen, palms not yellow. Cow ghee was applied to the cord. Mother HIV-negative at ANC; no BCG yet."

Working diagnosis: late-onset neonatal sepsis from **omphalitis with periumbilical flare** (staphylococcal risk), with hypoglycaemia and dehydration; jaundice probably physiological but worsened by sepsis and poor intake.

Worked example: plotting bilirubin. A term (39 weeks) baby at **36 hours** has TSB **240 umol/L**. On the NICE table, the phototherapy threshold at 36 h is **225 umol/L** and exchange **400**. So: **start phototherapy**, repeat TSB in 4-6 h, look for the cause (jaundice this high this early suggests haemolysis), and continue breastfeeding.

Differential diagnosis

The baby who is "not feeding well and sleepy":

Condition	Distinguishing features	Key investigation
Sepsis/meningitis	Risk factors, temperature instability, bulging fontanelle, mottling	Culture, CRP, LP
Hypoglycaemia	Jittery, sweaty, apnoea, seizures; SGA/IDM/preterm	Glucose
Hypothermia	Cold peripheries, temp below 36.5	Temperature
Severe jaundice/bilirubin encephalopathy	Deep jaundice, poor suck, then arching, high-pitched cry	TSB
Dehydration / breastfeeding failure	Weight loss above 10%, few wet nappies, hypernatraemia	Weight, sodium
HIE	History of asphyxia, onset from birth	Clinical
Congenital heart disease	Cyanosis, murmur, weak femoral pulses, hepatomegaly	SpO2 pre/post-ductal, echo
Neonatal tetanus	Day 3-14, could suck then stopped, trismus, spasms, alert between spasms	Clinical
Inborn error of metabolism	Well at birth then vomiting, encephalopathy after feeds	Ammonia, lactate, ketones

Jaundice by timing:

Onset	Likely causes
First 24 h (always pathological)	**Haemolysis** (ABO, Rh, G6PD, spherocytosis), sepsis, congenital infection (TORCH, syphilis)
Day 2-14	Physiological, breastfeeding (poor intake) jaundice, sepsis, haemolysis, bruising/cephalhaematoma, polycythaemia, prematurity
After 14 days (term)/21 days (preterm)	Breast milk jaundice (unconjugated, well baby), **UTI/sepsis**, **hypothyroidism**, G6PD; **conjugated**: **biliary atresia**, neonatal hepatitis, choledochal cyst

Investigations

Test	What you are looking for	Why
Glucose	Below 2.6 mmol/L	Treat immediately
Blood culture	Bacteraemia	Guides antibiotics
FBC	Infection, haemolysis, anaemia	Diagnosis and transfusion
CRP	Inflammation trend	Stopping antibiotics
CSF	Meningitis	Duration and drug choice
TSB, conjugated bilirubin	Severity and type of jaundice	Phototherapy/exchange; cholestasis
Blood group, DAT, G6PD, TSH	Cause of jaundice	Specific treatment and counselling
Swabs	Organism in eye/cord/skin	Targeted therapy

Management

1. Neonatal sepsis (possible serious bacterial infection)

Who to treat:

- Any **danger sign** (PSBI/very severe disease).
- **Risk factors** (ROM over 18 h, maternal fever above 38 C, foul liquor) in a baby under 1 day: **ampicillin + gentamicin for at least 2 days, then reassess** (WHO); stop if well and cultures negative.

Steps:

1. Stabilise (ABC, warmth, glucose, fluids/feeds).
2. **Blood culture** (and LP when stable) **before** the first dose if it causes no delay.
3. **Ampicillin + gentamicin** (first line, WHO 2013 and WHO 2024 SBI recommendations; UCG 2023).
4. **Supportive care**: warmth; feeds (EBM by NGT if unable to suck) or IV 10% dextrose; oxygen if SpO2 below 90%; no paracetamol for fever (undress, control environment).
5. **Review at 48-72 h**: improving → complete course; not improving → **change** (ceftriaxone, or per culture; cloxacillin if staphylococcal; meropenem at referral hospitals for resistant Gram-negatives).
6. **Duration**: stop at **48 h** if culture-negative and baby well (NICE NG195 allows stopping at **36 h** with negative culture and reassuring CRP); **7-10 days** for clinical/culture-proven sepsis (UCG 10 days); **21 days** for meningitis (UCG/WHO).

Antibiotic doses (WHO Pocket Book 2013; UCG 2023):

Drug	Dose	Route	Frequency	Duration / notes
Ampicillin	**50 mg/kg** (meningitis **100 mg/kg**)	IV/IM	**Every 12 h in week 1**; **every 8 h weeks 2-4** (UCG hospital protocol: 6-hourly IV)	With gentamicin; 7-10 days (21 days meningitis)
Gentamicin	**Week 1: 5 mg/kg** (normal weight), **3 mg/kg** LBW (WHO) / **4 mg/kg** preterm (UCG); **weeks 2-4: 7.5 mg/kg**	IV/IM	**Once daily** (24-hourly; longer intervals in very preterm or AKI)	Ototoxic/nephrotoxic; check levels where available

Drug	Dose	Route	Frequency	Duration / notes
Benzylpenicillin (HCII pre-referral or alternative to ampicillin)	**50,000 IU/kg**	IM/IV	Pre-referral single dose (UCG); 12-hourly week 1, 8-hourly after	Plus gentamicin
Cloxacillin (staphylococcal risk: omphalitis with flare, many pustules, abscess)	**50 mg/kg** (WHO range 25-50 mg/kg)	IV/IM	**Every 6 h** (UCG)	With **gentamicin 5-7 mg/kg once daily** (UCG)
Ceftriaxone (second line; meningitis)	**100 mg/kg once daily** (UCG); meningitis **50 mg/kg 12-hourly or 100 mg/kg once daily** (WHO)	IV/IM	Once (or twice) daily	**Avoid in hyperbilirubinaemia/preterm** if possible; **never with IV calcium**; cefotaxime 50 mg/kg 12-6-hourly is safer where available
Metronidazole (NEC, anaerobes)	7.5 mg/kg (some protocols 15 mg/kg load)	IV/oral	12-hourly week 1, 8-hourly after	With ampicillin + gentamicin for NEC

DRUG DOSING

Worked example: 3.0 kg baby, day 3 → ampicillin **150 mg IV 12-hourly** + gentamicin **15 mg IV once daily**. Same baby on day 10 → ampicillin **150 mg 8-hourly** + gentamicin **22.5 mg once daily** (7.5 mg/kg).

UGANDA CONTEXT

If referral is not possible (UCG/IMNCI), continue **ampicillin + gentamicin IM** for **7 days** at the health centre, keep warm, and support feeding. WHO's outpatient PSBI regimens (once-daily IM gentamicin plus oral amoxicillin for 7 days) are used for young infants whose families cannot accept referral, but never for babies with critical signs (unconscious, convulsions, not feeding at all, apnoea).

2. Neonatal jaundice

Assess every baby: look in daylight; **any jaundice in the first 24 h** → measure bilirubin urgently (NICE: within 2 h). Check palms/soles. Babies of 35 weeks or more after 24 h → TcB (or serum if no TcB).

Kramer's zones (rough guide only; always confirm high values):

Zone	Extent	Approx. bilirubin (umol/L)
1	Head and neck	About 100
2	Upper trunk (to umbilicus)	About 150
3	Lower trunk and thighs	About 200
4	Arms and lower legs	About 250
5	**Palms and soles**	**Above 250**

WHO Pocket Book thresholds (umol/L):

Age	Phototherapy: healthy 35 weeks or more	Phototherapy: below 35 weeks or any risk factor*	Exchange: healthy 35 weeks or more	Exchange: below 35 weeks or risk factor
Day 1	**Any visible jaundice**	Any visible jaundice	260	220

Age	Phototherapy: healthy 35 weeks or more	Phototherapy: below 35 weeks or any risk factor*	Exchange: healthy 35 weeks or more	Exchange: below 35 weeks or risk factor
Day 2	260	170	425	260
Day 3 or more	310	250	425	340

*Risk factors: small (below 2.5 kg or below 37 weeks), haemolysis, sepsis.

NICE CG98 (updated 2023) threshold table, 38 weeks or more (umol/L):

Age (h)	0	6	12	18	24	30	36	42	48	60	72	84	96+
Phototherapy	100	125	150	175	200	212	225	237	250	275	300	325	**350**
Exchange	100	150	200	250	300	350	400	450	450	450	450	450	**450**

For babies below 38 weeks use NICE's gestation-specific graphs (lower lines; rule of thumb for phototherapy after 72 h ≈ gestation in weeks × 10 – 100 umol/L). The **AAP 2022** guideline (USA) uses hour-specific curves by gestation (35 weeks or more) with **neurotoxicity risk factors** and an "**escalation of care**" threshold 34 umol/L (2 mg/dL) below the exchange line; many Ugandan units still use the WHO table or NICE charts.

Phototherapy: how to do it well

1. Baby **naked** (nappy only), **eyes covered**, under blue light (460-490 nm) at the right distance; turn 2-3-hourly or use double lights for intensive phototherapy.
2. **Continue breastfeeding** (short breaks, or EBM by cup); give **10-20% extra fluid** if under radiant heat or if dehydrated.
3. Monitor temperature, hydration, weight; **repeat TSB 4-6 h** after starting (rising fast or haemolysis), then 6-12-hourly.
4. **Stop** when TSB is **50 umol/L below the phototherapy threshold** (NICE); check for **rebound** 12-18 h later.
5. **Do not use sunlight** as treatment (unreliable, burns, hypothermia/hyperthermia).

Exchange transfusion (double volume = **160 mL/kg** in term babies; about 200 mL/kg in preterm) for TSB above the exchange threshold or **signs of acute bilirubin encephalopathy**, at a centre that can do it; keep intensive phototherapy going during transfer. **IVIG 500 mg/kg** over 4 h (NICE) for Rh/ABO haemolytic disease if TSB keeps rising above 8.5 umol/L per hour despite intensive phototherapy (AAP 2022: 0.5-1 g/kg if at escalation threshold with positive DAT).

Prolonged jaundice: check **conjugated bilirubin**, stool and urine colour, TSH, urine culture, G6PD; **pale stools or conjugated above 25 umol/L** → urgent referral (biliary atresia needs a Kasai procedure, ideally **before 60 days**). Breast milk jaundice: well, thriving baby with unconjugated hyperbilirubinaemia: **continue breastfeeding**.

3. Hypoglycaemia

At-risk babies (check glucose before the second feed, 2-4 h after birth, then before feeds): preterm, LBW/SGA, infant of a diabetic mother, asphyxia, sepsis, hypothermia, poor feeding, maternal beta-blockers (labetalol).

Situation	Treatment
Glucose **below 2.6**, **no symptoms**, baby can feed	**Breastfeed or give EBM** (e.g. 10 mL/kg by cup/NGT) now; recheck in **30-60 min**
Glucose **below 2.6** with **symptoms** (jittery, lethargic, apnoea, seizure) or not feeding, or persistent despite feeds	**10% dextrose 2 mL/kg IV** over 5 min, then **10% dextrose infusion** (60 mL/kg/day day 1 → GIR about 4-6 mg/kg/min); recheck in **30 min**
No IV access	**EBM or 10% dextrose by NGT** (Pocket Book), or sugar solution by cup (UCG HC level); refer

Situation	Treatment
Persistent despite **GIR above 8 mg/kg/min**	Think **hyperinsulinism** (IDM, SGA), adrenal insufficiency; refer; specialists may use glucagon or hydrocortisone

> **NOTE**
> Thresholds differ: Ugandan neonatal units commonly treat below **2.6 mmol/L** (used throughout these notes), WHO Pocket Book uses **2.2 mmol/L** for sick infants, and NICE/BAPM use **2.0 mmol/L** (with IV treatment for any value below 1.0 or symptomatic). Follow your unit's protocol and say which you used.

4. Hypothermia

1. **Prevent**: dry at birth, skin-to-skin, hat, **delay bathing 24 h**, warm room (at least 25 C), KMC for small babies.
2. **Cold stress (36.0-36.4)**: skin-to-skin, extra layers, cover head, warm room, breastfeed; recheck in 1 h.
3. **Moderate (32.0-35.9)**: KMC or incubator/radiant warmer (or a warm cot and room); check **glucose**, feed, look for sepsis; recheck every 30-60 min until normal.
4. **Severe (below 32)**: rapid rewarming in an incubator/radiant warmer, **IV 10% dextrose**, oxygen, **antibiotics**, monitor for apnoea and bradycardia.
5. Remove wet clothes; avoid hot-water bottles directly on skin (burns).

5. Cord, eye and skin infections

Condition	Features	Treatment (UCG 2023 / WHO)
Local umbilical infection	Red umbilicus or pus, redness **not** spreading	Clean with boiled cooled water, dry, **gentian violet 0.5%** 3 times daily; **oral amoxicillin** (UCG: 250 mg DT ¼ tab under 1 month/4 kg, ½ tab 1-2 months/4-6 kg, **12-hourly for 5 days**); review in 2 days
Omphalitis with periumbilical flare	Redness/swelling spreading onto the abdominal wall, pus, baby may be unwell	Admit; **cloxacillin 50 mg/kg IV 6-hourly + gentamicin**; blood culture; 7-10 days; watch for necrotising fasciitis and portal vein thrombosis
Cord bleeding	Oozing from the stump	Retie the cord; vitamin K if not given; refer if persists (UCG)
Sticky eye / mild conjunctivitis	Small amount of discharge, lids not swollen	Wash with clean water/breast milk; **tetracycline or chloramphenicol eye ointment 4 times daily for 5 days** (WHO); review 48 h
Gonococcal ophthalmia	**Day 2-5**, copious pus, **swollen lids**, risk of corneal perforation	**Ceftriaxone 125 mg IM once** (UCG; WHO 50 mg/kg, max 125 mg) **plus azithromycin 20 mg/kg orally daily for 3 days** (UCG, covers chlamydia); irrigate frequently with saline; admit; treat **both parents** for gonorrhoea and chlamydia and screen for HIV/syphilis
Chlamydial conjunctivitis	Day 5-14, watery then purulent, milder; may have pneumonia at 4-12 weeks	**Azithromycin 20 mg/kg daily for 3 days** (or oral erythromycin 50 mg/kg/day in 4 doses for 14 days); treat parents
Skin pustules, fewer than 10	Small pustules, baby well	Clean, **gentian violet 0.5%** 2-3 times daily; review in 2 days
Skin pustules 10 or more, bullae, abscess, cellulitis	Many/large pustules, swelling, hardness	Treat as **serious infection**: **cloxacillin + gentamicin** IV; drain abscess
Oral thrush	White plaques that do not wipe off	**Gentian violet** paint 4 times daily for 7 days (UCG) or **nystatin 100,000 IU (1 mL) 4 times daily for 7 days**; treat the mother's nipples

CLINICAL PEARL

Topical tetracycline at birth **prevents** ophthalmia neonatorum but **does not treat** established gonococcal infection; it needs systemic ceftriaxone (UCG: topical tetracycline has no added benefit in active disease).

6. Birth injuries

Injury	Recognition	Management
Caput succedaneum	Soft, pitting oedema over the presenting part, **crosses sutures**, present at birth	Reassure; resolves in **1-3 days**
Cephalhaematoma	Firm swelling **limited by suture lines** (usually parietal), appears hours after birth	Reassure; resolves in **weeks to 3 months**; **do not aspirate**; watch for **jaundice** and anaemia
Subgaleal haemorrhage	Boggy, fluctuant swelling **crossing sutures**, spreading to neck/forehead, **pallor, tachycardia, shock**, after **vacuum**	**Emergency**: saline 10 mL/kg, **blood**, vitamin K, FFP, monitor head circumference
Erb's palsy (C5-C6)	Arm adducted, internally rotated, elbow extended, wrist flexed ("**waiter's tip**"); asymmetric Moro; grasp intact	Gentle handling, pin sleeve to the vest in the first week, then **passive physiotherapy**; most recover in weeks to months; refer if no biceps recovery by **3 months**
Klumpke's palsy (C8-T1)	Claw hand, absent grasp, ± **Horner syndrome**	Physiotherapy; refer
Clavicle fracture	Crepitus, swelling, reduced arm movement, asymmetric Moro (often after shoulder dystocia)	Reassure; heals in **7-10 days** with callus; pin sleeve, gentle handling, analgesia
Humerus/femur fracture	Breech or difficult delivery; swelling, pain, no movement	X-ray; splint; orthopaedics
Facial nerve palsy	Asymmetric cry, eye does not close on the affected side (forceps)	Eye lubricant/protection; usually recovers
Bruising of buttocks/legs after breech	Bruises, swelling	Reassure (UCG): resolves in 1-2 weeks; watch jaundice; do not force the legs
Sternomastoid tumour	Firm lump in sternomastoid at 2-3 weeks, torticollis	Stretching physiotherapy

7. Respiratory distress and feeding problems (brief)

- **Respiratory distress** (RR 60 or more, indrawing, grunting): oxygen to SpO2 above 90% (90-95% in preterms), **antibiotics** (pneumonia/sepsis cannot be excluded), consider CPAP, look for **transient tachypnoea** (CS, mild, settles in 24-72 h), **meconium aspiration**, **RDS** (preterm), **pneumothorax**, **congenital heart disease** (cyanosis, murmur, weak femorals) and **hypothermia/hypoglycaemia**.
- **Feeding problems** (IMNCI): poor attachment, fewer than 8 feeds/24 h, other foods/drinks, thrush, low weight → correct positioning and attachment, feed more often day and night, stop other fluids, treat thrush, review in 2 days; refer for breastfeeding counselling if not improving.

Complications

Problem	Complications
Sepsis	Meningitis (hydrocephalus, deafness, CP, epilepsy), septic shock, DIC, NEC, osteomyelitis/septic arthritis, death

Problem	Complications
Jaundice	**Acute bilirubin encephalopathy → kernicterus** (athetoid CP, sensorineural deafness, upward gaze palsy, dental enamel dysplasia, intellectual impairment)
Hypoglycaemia	Seizures, occipital brain injury, visual impairment, developmental delay
Hypothermia	Hypoglycaemia, acidosis, apnoea, pulmonary haemorrhage, sclerema, death
Omphalitis	Necrotising fasciitis, sepsis, portal vein thrombosis → portal hypertension later
Gonococcal ophthalmia	Corneal ulceration, perforation, **blindness**
Birth injuries	Shock and death (subgaleal), permanent brachial plexus palsy

Ward-round & exam pearls

Model presentation

"**Baby Namusoke**, a **4-day-old** female, born at term by SVD at a HC IV, birth weight **3.0 kg**, today **2.75 kg (8% loss)**. Presented with **yellow eyes since day 2**, now spreading to the legs, and **poor feeding for 1 day**. Exclusively breastfed, **6 feeds in 24 h**, 3 wet nappies, yellow stools. Mother is **blood group O positive**, baby **B positive**, DAT not done. No fever, no family history of G6PD deficiency known.

On examination she is jaundiced to the **legs (Kramer 4)**, palms and soles not yellow, alert with a good cry, normal tone and suck, temperature 36.9 C, RR 46, no hepatosplenomegaly, umbilicus clean. TSB at **84 hours is 290 umol/L**, just **below** the NICE phototherapy line at 84 h (**325 umol/L**).

Assessment: neonatal jaundice, probably physiological/breastfeeding-related with possible **ABO incompatibility**; she is near the phototherapy threshold, and feeding is inadequate. **Plan:** support breastfeeding (8-12 feeds/day, EBM top-ups by cup), repeat TSB in 6-12 h, start phototherapy if above threshold, FBC and reticulocytes, DAT, G6PD if available, review danger signs.

Mother **HIV-negative** in labour. **BCG and OPV0 given** at birth. Weight-for-age Z-score about −1."

Viva questions (with answers)

1. **Name the WHO/UCG neonatal danger signs.** Not feeding well, convulsions, drowsy/unconscious or moves only when stimulated, RR 60 or more, severe chest indrawing, grunting, temperature above 37.5 or below 35.5 C, central cyanosis.
2. **First-line antibiotics for neonatal sepsis and doses?** Ampicillin 50 mg/kg 12-hourly (week 1)/8-hourly (weeks 2-4) + gentamicin 5 mg/kg (3-4 mg/kg LBW/preterm) once daily week 1, 7.5 mg/kg weeks 2-4.
3. **Why is jaundice on day 1 pathological?** Physiological processes cannot raise bilirubin that fast; it implies haemolysis or infection.
4. **How do you stop phototherapy?** When TSB is 50 umol/L below the threshold; check rebound in 12-18 h.
5. **What are the features of kernicterus?** Choreoathetoid CP, sensorineural deafness, upward gaze palsy, dental enamel dysplasia.
6. **How do you treat gonococcal ophthalmia?** Ceftriaxone 125 mg IM once + azithromycin 20 mg/kg daily × 3 days, saline washes, treat both parents.
7. **Caput vs cephalhaematoma?** Caput crosses sutures, present at birth, resolves in days; cephalhaematoma is limited by sutures, appears later, lasts weeks.

OSCE tips

- **Counsel a mother on danger signs and cord care** (chlorhexidine 7 days, nothing else on the stump, keep dry, fold nappy below).
- **Explain phototherapy** to a mother: why, eye pads, keep breastfeeding, temperature checks, how long.
- **Prescribe** antibiotics for a given weight and age; show mg/kg and the total dose.

Mnemonics & memory aids

Danger signs: "FEEDS-COLD"

- **F**eeding poorly
- **E**pisodes of convulsion
- **E**ffortful breathing (grunting, severe indrawing)
- **D**rowsy / moves only when stimulated
- **S**peed: RR 60 or more
- **C**yanosis
- **O**ver 37.5 C
- **L**ow temperature below 35.5 C
- **D**on't wait: treat and refer

Jaundice timing: "1-14-21": day **1** = always pathological; beyond **14** days (term) or **21** days (preterm) = prolonged → check conjugated.

Head swellings: "Caput Crosses, Cephal Confined, Subgaleal Shocks."

High-yield summary

Numbers & doses to know

Item	Value
Neonatal period	**28 days**
EOS / LOS	**Below 72 h** / after 72 h
Sepsis risk factors	**ROM over 18 h**, maternal fever **above 38 C**, foul liquor
Ampicillin	**50 mg/kg** IV/IM **12-hourly week 1, 8-hourly weeks 2-4** (100 mg/kg meningitis)
Gentamicin	**5 mg/kg** daily week 1 (**3 mg/kg** LBW WHO; **4 mg/kg** preterm UCG); **7.5 mg/kg** weeks 2-4
Cloxacillin	**50 mg/kg 6-hourly** + gentamicin (staphylococcal)
Ceftriaxone	**100 mg/kg once daily** (2nd line); avoid with jaundice/IV calcium
Course length	Stop at **48 h** if well and culture negative; sepsis **7-10 days**; meningitis **21 days**
Jaundice	Day 1 = pathological; TSB within **2 h**
NICE phototherapy (38 weeks or more)	**200 at 24 h**, **250 at 48 h**, **300 at 72 h**, **350 at 96 h+**
NICE exchange (38 weeks or more)	**300 at 24 h**, **450 from 42 h**
WHO phototherapy (healthy 35 weeks or more)	Day 1 **any visible**, day 2 **260**, day 3+ **310**
Stop phototherapy	**50 umol/L below threshold**

Item	Value
Exchange volume	**160 mL/kg** (double volume, term)
IVIG (isoimmune)	**500 mg/kg** over 4 h
Conjugated hyperbilirubinaemia	**Above 25 umol/L**; Kasai **before 60 days**
Hypoglycaemia	**Below 2.6 mmol/L → 10% dextrose 2 mL/kg** IV, recheck in 30 min
Hypothermia	**36.5-37.5** normal; **below 32** severe
Gonococcal ophthalmia	**Ceftriaxone 125 mg IM once + azithromycin 20 mg/kg × 3 days**
Cord care	**Chlorhexidine 7.1% daily for 7 days**
Local skin infection	**Fewer than 10 pustules** → gentian violet; 10 or more → IV cloxacillin + gentamicin

Classic exam traps

- **"Jaundice at 12 hours: it's physiological, review tomorrow."** → Wrong. Day-1 jaundice is **always pathological**; measure bilirubin now.
- **"Put the jaundiced baby in the morning sun."** → Wrong. Sunlight is unreliable and unsafe; use **phototherapy**.
- **"Stop breastfeeding for jaundice."** → Wrong. **Continue breastfeeding**; poor intake worsens jaundice.
- **"Febrile neonate: give paracetamol and observe."** → Wrong. Treat as **sepsis**; control the environment; no antipyretics (UCG).
- **"Gentamicin 7.5 mg/kg on day 2 of life."** → Too high: **5 mg/kg** in week 1 (3-4 mg/kg if LBW/preterm); 7.5 mg/kg is for **weeks 2-4**.
- **"Ceftriaxone for a deeply jaundiced preterm with sepsis."** → Avoid if possible (displaces bilirubin); use ampicillin + gentamicin or cefotaxime.
- **"Gonococcal ophthalmia: tetracycline ointment is enough."** → Wrong. Needs **systemic ceftriaxone** (and treat parents).
- **"Aspirate the cephalhaematoma."** → Wrong. Leave it; it resolves (aspiration risks infection).

Questions seniors ask

1. **"What is the baby's age in hours, and where is the bilirubin on the chart?"** Give the TSB, the threshold for that age/gestation, and the decision.
2. **"Why is this baby on antibiotics, and when will you stop?"** Name the risk factor/sign, the culture result plan and the duration.
3. **"What dose of gentamicin, and why that dose?"** mg/kg by age and weight band; check renal function.
4. **"Is this jaundice conjugated or unconjugated? What colour are the stools?"** Pale stools = urgent biliary atresia work-up.
5. **"What is the glucose, and how often are you checking it?"** Value, threshold, frequency, treatment given.
6. **"Is this omphalitis local or spreading?"** Flare onto the abdominal wall changes oral to IV treatment.
7. **"What will you tell the mother before discharge?"** Danger signs, cord care, exclusive breastfeeding, warmth, return for jaundice of palms/soles, immunisation at 6 weeks.

SELF-TEST

Flashcards

27 cards. Cover the right-hand column and test yourself.

#	Question	Answer
1	List the WHO/UCG danger signs in a newborn or young infant.	**Not feeding well**, **convulsions**, **drowsy/unconscious or moves only when stimulated**, **RR 60 or more**, **severe chest indrawing**, **grunting**, temperature **above 37.5 C or below 35.5 C**, **central cyanosis**.
2	Define early- and late-onset neonatal sepsis.	**Early-onset**: first **72 h** (some use 7 days), usually maternal organisms acquired in labour. **Late-onset**: after 72 h, community or hospital-acquired.
3	Maternal risk factors that justify antibiotics in a newborn under 1 day (WHO).	**ROM more than 18 h** before delivery, maternal **fever above 38 C** in labour, **foul-smelling or purulent liquor**. Give ampicillin + gentamicin for at least 2 days and reassess.
4	Ampicillin dose and frequency in neonatal sepsis (WHO).	**50 mg/kg IV/IM every 12 h in week 1, every 8 h in weeks 2-4**; **100 mg/kg/dose** for meningitis. (UCG hospital protocol: 50 mg/kg IV 6-hourly.)
5	Gentamicin dose in neonates (WHO/UCG).	Week 1: **5 mg/kg once daily** (normal weight), **3 mg/kg** LBW (WHO) or **4 mg/kg** preterm (UCG). Weeks 2-4: **7.5 mg/kg once daily**.
6	Antibiotics when staphylococcal infection is likely (UCG 2023).	**Cloxacillin 50 mg/kg IV/IM 6-hourly + gentamicin 5-7 mg/kg once daily**; for omphalitis with flare, many pustules or abscess.
7	Second-line antibiotic if no improvement after 48-72 h (UCG).	Change ampicillin to **ceftriaxone 100 mg/kg once daily** (or per culture). Avoid ceftriaxone in jaundiced/preterm babies if possible and never with IV calcium.
8	Duration of antibiotics in neonatal infection.	Culture negative and well: **stop at 48 h** (NICE: 36 h with reassuring CRP). Sepsis **7-10 days** (UCG 10). Meningitis **21 days**.
9	Why is jaundice in the first 24 hours always pathological?	Physiological mechanisms cannot raise bilirubin that fast; it means **haemolysis** (ABO, Rh, G6PD, spherocytosis) or **infection**. Measure TSB within **2 h** (NICE).
10	IMNCI/UCG definition of severe jaundice in a young infant.	**Any jaundice in a baby under 24 h old**, or **yellow palms and soles at any age** → refer urgently, prevent hypoglycaemia, keep warm.
11	Define prolonged jaundice and what to check.	Jaundice beyond **14 days (term)** or **21 days (preterm)**. Check **conjugated bilirubin** (above 25 umol/L abnormal), **stool/urine colour**, TSH, urine culture, G6PD.
12	NICE phototherapy thresholds for a baby of 38 weeks or more.	**100** at birth, **200 at 24 h**, **250 at 48 h**, **300 at 72 h**, **350 umol/L from 96 h**.
13	NICE exchange thresholds for a baby of 38 weeks or more.	**100** at birth, **300 at 24 h**, rising to **450 umol/L from 42 h** onwards, or any signs of acute bilirubin encephalopathy.
14	WHO Pocket Book phototherapy thresholds (healthy 35 weeks or more).	Day 1: **any visible jaundice**; day 2: **260 umol/L**; day 3+: **310 umol/L**. Lower for preterm/risk factors (day 2 170, day 3+ 250).
15	When do you stop phototherapy and what then?	When TSB is **50 umol/L below the phototherapy threshold** (NICE); recheck for **rebound 12-18 h** later.
16	Features of acute bilirubin encephalopathy and kernicterus.	Acute: lethargy, poor suck, hypotonia → **hypertonia, retrocollis, opisthotonus, high-pitched cry**, fever, seizures. Chronic (kernicterus): **choreoathetoid CP**, **sensorineural deafness**, upward gaze palsy, dental enamel dysplasia.

#	Question	Answer
17	Kramer zones of jaundice.	1 head/neck (~100), 2 upper trunk (~150), 3 lower trunk/thighs (~200), 4 arms/lower legs (~250), 5 **palms and soles (above 250 umol/L)**. A rough guide only.
18	Treatment of symptomatic neonatal hypoglycaemia.	**10% dextrose 2 mL/kg IV** over 5 min, then **10% dextrose infusion** (60 mL/kg/day day 1); recheck glucose in **30 min**. No IV: EBM or 10% dextrose by NGT.
19	Babies at risk of hypoglycaemia.	**Preterm**, **LBW/SGA**, **infant of a diabetic mother**, **asphyxia**, **sepsis**, **hypothermia**, delayed/poor feeding, maternal beta-blockers.
20	Local umbilical infection: treatment per UCG.	Clean with boiled cooled water, dry, **gentian violet 0.5% three times daily**, **oral amoxicillin** (250 mg DT: ¼ tab under 1 month/4 kg, 12-hourly for 5 days), **review in 2 days**.
21	Treatment of gonococcal ophthalmia neonatorum (UCG 2023).	**Ceftriaxone 125 mg IM single dose + azithromycin syrup 20 mg/kg once daily for 3 days**, frequent saline irrigation, admit/follow up in 2 days, **treat both parents** and screen for HIV/syphilis.
22	Caput succedaneum vs cephalhaematoma vs subgaleal haemorrhage.	**Caput**: soft oedema, crosses sutures, present at birth, gone in days. **Cephalhaematoma**: limited by sutures, appears later, lasts weeks, jaundice. **Subgaleal**: boggy, crosses sutures, spreads, causes **shock** (vacuum).
23	Describe Erb's palsy.	**C5-C6** injury: arm adducted, internally rotated, elbow extended, wrist flexed (**waiter's tip**), absent Moro on that side, **grasp preserved**. Most recover; refer if no biceps recovery by 3 months.
24	Skin pustules: when local vs systemic treatment?	**Fewer than 10** pustules, baby well: clean + **gentian violet**. **10 or more**, bullae, abscess or swelling/hardness: **serious infection** → IV cloxacillin + gentamicin.
25	WHO classification of hypothermia.	Normal **36.5-37.5 C**; cold stress **36.0-36.4**; moderate **32.0-35.9**; severe **below 32 C**.
26	Double-volume exchange transfusion volume.	**2 × 80 = 160 mL/kg** in term babies (about 200 mL/kg in preterm); keep intensive phototherapy going.
27	Why avoid ceftriaxone in jaundiced neonates?	It **displaces bilirubin from albumin**, raising free bilirubin and kernicterus risk; it also precipitates with **IV calcium** (fatal reactions). Use cefotaxime where available.

SELF-TEST

Single best answer questions

15 questions. The answers and explanations follow at the end.

1. A 5-day-old, 3.0 kg baby is brought to a HC IV with poor feeding for 1 day. Temperature 35.3 C after warming, RR 66/min, mild chest indrawing, glucose 3.1 mmol/L. The referral hospital is 1 hour away. What is the MOST appropriate action?

A Give oral amoxicillin and review in 2 days

B Give paracetamol and reassess in 4 hours

C Give ampicillin 150 mg IM + gentamicin 15 mg IM, keep warm (skin-to-skin), support feeding, and refer urgently

D Admit and observe without antibiotics until a blood culture is available

E Give ceftriaxone 300 mg IM and send home

2. A term baby (3.4 kg) on day 12 is admitted with fever 38.5 C, lethargy and a bulging fontanelle. CSF shows 450 white cells/mm3. Which regimen is MOST appropriate?

A Ampicillin 340 mg IV 8-hourly plus gentamicin 25.5 mg IV once daily for 21 days

B Ampicillin 170 mg IV 12-hourly plus gentamicin 17 mg IV once daily for 7 days

C Oral amoxicillin for 10 days

D Benzylpenicillin alone for 5 days

E Ceftriaxone single dose IM

3. A baby born at 39 weeks is noticed to be yellow at 14 hours of age. She is feeding well. Mother is group O positive. What is the MOST appropriate next step?

A Reassure; review at the 6-day postnatal visit

B Advise sunlight exposure for 30 minutes twice daily

C Stop breastfeeding and give glucose water

D Measure serum bilirubin urgently (within 2 hours), check baby's blood group and DAT, and plot against the age-specific threshold

E Start ceftriaxone for presumed sepsis

4. A term baby at 48 hours has a TSB of 270 umol/L. He is well, feeding and not haemolysing. According to NICE CG98 (38 weeks or more), what is the MOST appropriate management?

A No treatment; recheck in 1 week

B Exchange transfusion

C IVIG 500 mg/kg

D Phenobarbital to induce liver enzymes

E Start phototherapy and repeat TSB in 6-12 hours

5. A 7-week-old exclusively breastfed baby is still jaundiced. He is thriving, but his stools are pale cream and his urine stains the nappy dark yellow. What is the MOST important next step?

A Reassure; this is breast milk jaundice

B Measure conjugated bilirubin and refer urgently to a paediatric surgeon/hepatology service for possible biliary atresia

C Start phototherapy

D Stop breastfeeding for 48 hours

E Give vitamin K and review in 1 month

6. A 6-hour-old infant of a mother with gestational diabetes (birth weight 4.3 kg) is jittery and feeding poorly. Glucose is 1.8 mmol/L. IV access is obtained. What is the MOST appropriate treatment?

A 10% dextrose 8.6 mL IV over 5 minutes, then a 10% dextrose infusion, recheck glucose in 30 minutes
B 50% dextrose 2 mL/kg IV
C Formula feed only and recheck in 4 hours
D Glucagon as first line
E Hydrocortisone 10 mg IV

7. A 4-day-old baby has a red umbilicus with pus. The redness extends 2 cm onto the surrounding abdominal wall. He is feeding but less vigorously; temperature 37.9 C. What is the MOST appropriate management?

A Gentian violet and review in 2 days
B Apply ash or herbal paste to dry the stump
C Oral amoxicillin for 5 days at home
D Admit, blood culture, cloxacillin 50 mg/kg IV 6-hourly plus gentamicin
E Chlorhexidine gel only

8. A 3-day-old baby has markedly swollen eyelids with copious thick yellow pus from both eyes. The mother had vaginal discharge in pregnancy that was not treated. Which treatment is MOST appropriate according to UCG 2023?

A Tetracycline eye ointment alone 3 times daily
B Chloramphenicol eye drops and review in 1 week
C Ceftriaxone 125 mg IM once plus azithromycin 20 mg/kg daily for 3 days, saline eye washes, and treatment of both parents
D Breast milk drops into the eyes
E Oral amoxicillin for 7 days

9. After a vacuum delivery, a baby has a firm swelling over the right parietal bone that does not cross the suture lines, first noticed at 8 hours. He is well, Hb normal. What is the MOST appropriate advice?

A Aspirate the swelling to prevent calcification
B This is a cephalhaematoma: reassure, it resolves over weeks; watch for jaundice
C This is a subgaleal haemorrhage: transfuse now
D This is caput succedaneum: it will be gone by tomorrow
E Apply a pressure bandage

10. A baby born after shoulder dystocia holds the right arm adducted and internally rotated, with the elbow extended and the wrist flexed. The grasp reflex is present. What is the MOST likely diagnosis?

A Klumpke palsy (C8-T1)
B Fractured humerus
C Cerebral palsy
D Erb's palsy (C5-C6)
E Septic arthritis of the shoulder

11. A 2-day-old, 2.2 kg baby born at 36 weeks has an axillary temperature of 34.8 C in the postnatal ward. What is the BEST immediate management?

A Give a warm bath
B Put a hot-water bottle directly against the skin
C Start antibiotics only
D Wrap in a single blanket and recheck in 6 hours
E Skin-to-skin KMC or incubator, cover the head, check glucose and feed, look for signs of sepsis, recheck temperature within an hour

12. A 3.2 kg baby born to a mother with ROM for 24 hours and fever 38.6 C in labour is well at 1 hour. What is the MOST appropriate plan?

A Take a blood culture (if available) and start ampicillin + gentamicin; reassess at 48 hours and stop if well and culture negative
B Observe only; antibiotics only if the baby becomes unwell
C Give a 10-day course of ceftriaxone
D Give oral amoxicillin for 5 days
E Treat the mother only

13. A term baby on phototherapy for jaundice (no haemolysis) has a TSB of 205 umol/L at 96 hours. The NICE phototherapy threshold at this age is 350 umol/L. What is the MOST appropriate decision?

A Continue phototherapy until the baby is no longer visibly yellow
B Start exchange transfusion
C Stop phototherapy (more than 50 umol/L below the threshold) and recheck bilirubin 12-18 hours later for rebound
D Stop and discharge with no further checks
E Add IVIG

14. A 2-day-old baby has 4 small pustules on the neck and axilla. He is feeding well and active, temperature 36.9 C. What is the MOST appropriate management per UCG 2023?

A Admit for IV cloxacillin and gentamicin
B Clean with boiled cooled water, dry, apply gentian violet 0.5%, and review in 2 days
C Apply hydrocortisone cream
D Bathe the baby in salty water
E No treatment needed

15. A 10-day-old preterm baby (1.6 kg) is septic and deeply jaundiced (TSB near the exchange threshold). Blood culture is pending. Which antibiotic choice is MOST appropriate to AVOID if an alternative exists?

A Ampicillin
B Gentamicin
C Cloxacillin
D Ceftriaxone
E Cefotaxime

Answers & explanations

1. C · Give ampicillin 150 mg IM + gentamicin 15 mg IM, keep warm (skin-to-skin), support feeding, and refer urgently

C is right: hypothermia below 35.5 C after warming, fast breathing and poor feeding are danger signs (very severe disease/PSBI). UCG/IMNCI: give the first dose of ampicillin 50 mg/kg (150 mg) plus gentamicin 5 mg/kg (15 mg, under 7 days) IM, prevent hypoglycaemia, keep warm and refer urgently. A is wrong: oral amoxicillin is for local bacterial infection only. B is wrong: she is hypothermic, and antipyretics are not used in neonates. D is wrong: blood culture is not available at HC IV and antibiotics must not be delayed. E is wrong: a baby with danger signs needs admission, and ceftriaxone is second line.

2. A · Ampicillin 340 mg IV 8-hourly plus gentamicin 25.5 mg IV once daily for 21 days

A is right: this is neonatal meningitis in week 2. Ampicillin meningitis dose is 100 mg/kg (340 mg) and from week 2 is given 8-hourly; gentamicin in weeks 2-4 is 7.5 mg/kg once daily (25.5 mg); meningitis needs 21 days (UCG/WHO). B is wrong: it uses sepsis doses, week-1 frequency and too short a course. C is wrong: oral therapy is inadequate for meningitis. D is wrong: monotherapy for 5 days is inadequate for Gram-negative and other neonatal pathogens. E is wrong: a single dose treats ophthalmia, not meningitis.

3. D · Measure serum bilirubin urgently (within 2 hours), check baby's blood group and DAT, and plot against the age-specific threshold

D is right: jaundice in the first 24 h is always pathological (NICE: TSB within 2 h); a group O mother raises the possibility of ABO haemolytic disease, so check the baby's group and DAT, and use the hour-specific chart (NICE phototherapy at 12 h is 150 umol/L). A is wrong: day-1 jaundice needs same-day assessment. B is wrong: sunlight is not a safe or reliable treatment. C is wrong: stopping breastfeeding worsens jaundice and dehydration. E is wrong: sepsis is possible, but ceftriaxone is a poor choice in a jaundiced neonate and there are no other signs of sepsis yet.

4. E · Start phototherapy and repeat TSB in 6-12 hours

E is right: the NICE phototherapy threshold at 48 h is 250 umol/L and the exchange threshold is 450, so 270 needs phototherapy with repeat TSB. A is wrong: the level is above the phototherapy line. B is wrong: it is far below the exchange threshold and there are no signs of encephalopathy. C is wrong: IVIG is for isoimmune haemolysis with rapidly rising bilirubin. D is wrong: phenobarbital is not recommended for treating neonatal jaundice.

5. B · Measure conjugated bilirubin and refer urgently to a paediatric surgeon/hepatology service for possible biliary atresia

B is right: pale stools and dark urine in prolonged jaundice indicate conjugated (cholestatic) jaundice; biliary atresia must be excluded urgently because the Kasai operation works best before about 60 days. A is wrong: breast milk jaundice is unconjugated with normal yellow stools. C is wrong: phototherapy does not treat conjugated jaundice (and can cause bronze baby syndrome). D is wrong: stopping breastfeeding is unnecessary and harmful. E is wrong: vitamin K may be needed (fat malabsorption), but delaying a month risks liver failure.

6. A · 10% dextrose 8.6 mL IV over 5 minutes, then a 10% dextrose infusion, recheck glucose in 30 minutes

A is right: symptomatic hypoglycaemia (jittery, poor feeding, 1.8 mmol/L) needs 10% dextrose 2 mL/kg IV (4.3 kg × 2 = 8.6 mL), followed by an infusion and a recheck in 30 min; IDM babies have hyperinsulinism and often need a continuous infusion. B is wrong: 50% dextrose is hyperosmolar, damages veins and causes rebound hypoglycaemia in neonates. C is wrong: a symptomatic baby who is feeding poorly needs IV glucose, and 4 hours is too long. D and E are wrong: glucagon and hydrocortisone are for persistent hypoglycaemia despite a high glucose infusion rate, under specialist advice.

7. D · Admit, blood culture, cloxacillin 50 mg/kg IV 6-hourly plus gentamicin

D is right: redness spreading onto the abdominal wall (periumbilical flare) with fever is serious bacterial infection with staphylococcal risk; UCG advises cloxacillin 50 mg/kg 6-hourly plus gentamicin, after cultures. A and C are wrong: gentian violet and oral amoxicillin are for local umbilical infection without spreading redness or systemic signs. B is wrong: applying substances to the stump is harmful (infection, tetanus). E is wrong: chlorhexidine is for prevention, not treatment of established cellulitis.

8. C · Ceftriaxone 125 mg IM once plus azithromycin 20 mg/kg daily for 3 days, saline eye washes, and treatment of both parents

C is right: copious purulent discharge with lid swelling on day 2-5 suggests gonococcal ophthalmia, which can perforate the cornea; UCG treats both gonorrhoea and chlamydia with ceftriaxone 125 mg IM stat plus azithromycin 20 mg/kg daily for 3 days, with irrigation and treatment of both parents. A is wrong: UCG states topical tetracycline has no added benefit in active disease. B is wrong: topical therapy alone and a week's delay risk blindness. D is wrong: breast milk is not a treatment for gonococcal infection. E is wrong: amoxicillin does not reliably treat gonococcus (resistance) or chlamydia.

9. B · This is a cephalhaematoma: reassure, it resolves over weeks; watch for jaundice

B is right: a firm swelling limited by suture lines, appearing hours after birth, is a subperiosteal cephalhaematoma; it resolves over weeks to 3 months, and breakdown of the blood can increase jaundice. A is wrong: aspiration risks infection. C is wrong: subgaleal haemorrhage is boggy, crosses sutures and causes shock. D is wrong: caput is oedema present at birth that crosses sutures. E is wrong: pressure bandaging is unnecessary.

10. D · Erb's palsy (C5-C6)

D is right: the 'waiter's tip' posture with preserved grasp is Erb's palsy from stretching of the upper brachial plexus roots (C5-C6), typically after shoulder dystocia. A is wrong: Klumpke palsy affects the hand (claw hand, absent grasp, possibly Horner syndrome). B is wrong: a fracture causes swelling, pain and crepitus; it can coexist but does not give this posture with intact grasp. C is wrong: cerebral palsy is not diagnosed at birth and does not cause an isolated flaccid arm. E is wrong: septic arthritis develops days later with fever and swelling.

11. E · Skin-to-skin KMC or incubator, cover the head, check glucose and feed, look for signs of sepsis, recheck temperature within an hour

E is right: 34.8 C is moderate hypothermia (32.0-35.9 C). Rewarm with skin-to-skin/KMC or an incubator, cover the head, check and treat hypoglycaemia, feed, and look for sepsis; recheck within the hour. A is wrong: bathing increases evaporative heat loss. B is wrong: hot-water bottles directly on skin cause burns. C is wrong: antibiotics may be needed if sepsis is suspected, but rewarming and glucose come first. D is wrong: a single blanket and a 6-hour wait are inadequate.

12. A · Take a blood culture (if available) and start ampicillin + gentamicin; reassess at 48 hours and stop if well and culture negative

A is right: WHO advises IM/IV ampicillin + gentamicin for at least 2 days for newborns with documented risk factors (ROM over 18 h, maternal fever above 38 C, foul liquor), continuing only if there are signs of sepsis or a positive culture. B is wrong: in a setting with limited monitoring, both risk factors together justify prophylaxis/empirical treatment. C is wrong: ceftriaxone is second line and a 10-day course is unnecessary if the baby stays well with negative cultures. D is wrong: oral amoxicillin is not adequate empirical therapy for early-onset sepsis. E is wrong: the baby is at risk too.

13. C · Stop phototherapy (more than 50 umol/L below the threshold) and recheck bilirubin 12-18 hours later for rebound

C is right: NICE advises stopping phototherapy once TSB is 50 umol/L or more below the phototherapy threshold, and checking for rebound 12-18 h after stopping. A is wrong: visible jaundice persists below treatment thresholds; continuing exposes the baby to unnecessary separation and fluid loss. B is wrong: the level is far below the exchange line. D is wrong: a rebound check is recommended. E is wrong: there is no haemolysis and the level is falling.

14. B · Clean with boiled cooled water, dry, apply gentian violet 0.5%, and review in 2 days

B is right: fewer than 10 pustules in a well baby is a local skin infection; UCG advises home treatment (wash, dry, gentian violet 0.5%) and review in 2 days. A is wrong: systemic antibiotics are for 10 or more pustules, bullae, swelling/hardness or a sick baby. C is wrong: steroids worsen bacterial infection. D is wrong: salty baths are not a treatment and irritate the skin. E is wrong: these are pustules, not benign erythema toxicum, and need treatment and review.

15. D · Ceftriaxone

D is right: ceftriaxone displaces bilirubin from albumin, increasing the risk of kernicterus in a deeply jaundiced preterm, and must never be given with IV calcium in neonates. A, B and C are wrong as answers because ampicillin, gentamicin and cloxacillin do not displace bilirubin and are standard neonatal choices. E is wrong as an answer because cefotaxime is the preferred third-generation cephalosporin in jaundiced neonates.

SELF-TEST

Clinical cases

Work through each step before reading the model answer.

Case 1: The cold, sleepy baby with a smelly cord

A 7-day-old boy is brought to Mengo emergency by his grandmother. He was born at home in Mukono to a 17-year-old mother; birth weight was not measured. Cow ghee was applied to the cord. For 2 days he has been feeding poorly and 'sleeping too much'. Weight 2.6 kg, axillary temperature 35.2 C, RR 68/min with chest indrawing, HR 178, CRT 4 seconds. The umbilicus is red and smelly with redness spreading 3 cm onto the abdomen.

1. What are your first actions?

> **ABC**: oxygen if SpO2 below 90%; **IV/IO access**; **glucose** check; warm (incubator/radiant warmer or skin-to-skin once stable); CRT 4 s with HR 178 = poor perfusion → **normal saline 10 mL/kg = 26 mL** over 10-20 min and reassess; blood culture; start antibiotics immediately.

2. Glucose is 1.9 mmol/L. What do you give?

> **10% dextrose 2 mL/kg = 5.2 mL IV** over 5 min, then **10% dextrose maintenance** (about 60-90 mL/kg/day by day 7 depending on hydration, with sodium after day 2, e.g. 10% dextrose/0.18% saline); recheck glucose in 30 min.

3. Which antibiotics and doses?

> Omphalitis with flare = staphylococcal risk: **cloxacillin 50 mg/kg = 130 mg IV 6-hourly + gentamicin** (day 7: 5 mg/kg = 13 mg once daily per UCG/WHO week 1; from day 8, 7.5 mg/kg = 19.5 mg). Add cover for Gram-negatives/anaerobes if necrotising fasciitis suspected; LP once stable.

4. What other diagnosis must you think of in a home birth with substances on the cord?

> **Neonatal tetanus** (day 3-14, inability to suck, trismus, spasms). Check for trismus, rigidity, spasms on stimulation; ask the mother's Td status.

5. On day 3 of treatment, he is warm, feeding by NGT, culture grows S. aureus sensitive to cloxacillin. Plan?

> Continue **cloxacillin** (stop gentamicin if sensitivities allow) for **10-14 days** total (longer if deep infection/osteomyelitis); daily cord care; advance EBM to cup/breast; daily weight; **BCG/OPV0** before discharge; counsel on cord care (chlorhexidine, nothing else).

6. What counselling and public health actions are needed?

> Teach the mother and grandmother **clean cord care**, hand-washing, exclusive breastfeeding, warmth, danger signs; check and start **maternal Td**; HIV and syphilis testing for the mother if not done; family planning for the teenage mother; link to the village health team for follow-up.

Take-away: A cold, sleepy neonate with a spreading umbilical infection has sepsis with shock risk: give oxygen, fluid, glucose and anti-staphylococcal antibiotics in mg/kg doses at once, and think of tetanus when the cord has been contaminated.

Case 2: Early jaundice in a group O mother's baby

At Kawempe National Referral Hospital, a 39-week, 3.3 kg baby girl born by SVD is noticed to be yellow at 18 hours of age. The mother is a para 2, blood group O positive; her first child 'needed blue lights'. The baby is feeding well. Temperature 36.8 C.

1. Why is this jaundice concerning?

> **Jaundice under 24 h is always pathological.** A group O mother and a sibling who needed phototherapy suggest **ABO haemolytic disease**; also consider G6PD deficiency, Rh disease (mother is Rh positive, so unlikely), sepsis and congenital infection.

2. What investigations do you request?

> **TSB now** (NICE: within 2 h), **baby's blood group and DAT**, FBC with reticulocytes and film (spherocytes), G6PD if available; blood culture and CRP if any sign of infection; repeat TSB 4-6-hourly while rising.

3. TSB at 18 h is 190 umol/L. Interpret it.

> NICE (38 weeks or more): phototherapy threshold at 18 h = **175 umol/L**, exchange = **250 umol/L**. She is **above the phototherapy line**: start phototherapy now (intensive/double lights if rising fast), continue breastfeeding, repeat TSB in 4-6 h.

4. Six hours later (24 h) TSB is 260 umol/L despite phototherapy. DAT is positive. What next?

> Rise of 70 in 6 h = **11.7 umol/L/h** (above 8.5/h). The 24-h exchange threshold is **300**. Give **intensive multiple phototherapy**, consider **IVIG 500 mg/kg (1.65 g) over 4 h** (NICE, isoimmune haemolysis with rise above 8.5 umol/L/h), cross-match **O-negative/compatible blood** for possible **double-volume exchange (160 mL/kg = about 530 mL)**, recheck TSB in 4 h, watch for encephalopathy.

5. What are the early signs of acute bilirubin encephalopathy you will monitor for?

> **Lethargy, poor suck, hypotonia**, then **high-pitched cry, hypertonia, retrocollis and opisthotonus**, fever, seizures. Any of these = **immediate exchange transfusion**.

6. What follow-up does she need?

> Rebound TSB after stopping phototherapy; **Hb at 2-4 weeks** (late anaemia from ongoing haemolysis); **hearing screen**; development review; counsel about risk in future pregnancies (ABO disease often recurs).

Take-away: Day-1 jaundice is haemolysis or sepsis until proven otherwise: measure TSB, plot it by age in hours, start phototherapy early, and escalate to IVIG or exchange when the rise is fast or near the exchange line.

Case 3: Sticky eyes and a sick mother

A 4-day-old baby born at a HC III in Luwero is brought to a HC IV because both eyes are swollen shut with thick yellow pus since yesterday. No tetracycline was given at birth. The mother had an untreated vaginal discharge in the third trimester; her HIV test at ANC was negative but she has not been retested. The baby is feeding well, temperature 37.0 C, weight 3.1 kg.

1. What is the likely diagnosis and why is it urgent?

> **Gonococcal ophthalmia neonatorum** (onset day 2-5, copious pus, lid oedema). It can cause **corneal ulceration and perforation within days → blindness**.

2. How do you treat the baby?

> **Ceftriaxone 125 mg IM single dose** (WHO 50 mg/kg, max 125 mg) **plus azithromycin 20 mg/kg = 62 mg orally once daily for 3 days** (UCG, chlamydia co-infection); **irrigate eyes with saline** frequently (gloves, open lids carefully: pus may be under pressure); admit or review twice daily; refer urgently if no improvement or corneal haze.

3. What investigations would help where available?

> **Eye swab Gram stain** (Gram-negative intracellular diplococci) and culture; check for systemic gonococcal infection (sepsis, arthritis) if unwell.

4. What must be done for the parents?

> Treat **both parents** for gonorrhoea and chlamydia per UCG STI guidelines; **retest the mother for HIV and syphilis** (and HBsAg); partner notification; condom counselling.

5. How could this have been prevented?

> **ANC STI screening and treatment**, and **tetracycline 1% eye ointment to both eyes of all newborns at delivery** (UCG prevention).

Take-away: Copious purulent eye discharge in the first week is gonococcal until proven otherwise: give systemic ceftriaxone plus azithromycin, irrigate, and treat both parents.

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A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.